

Applied Statistics for Engineers and Scientists II - 23TT2

Chapter 01: Multiple linear regression

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Important note: All of these programming exercises are solved using the  programming language. For a quick introduction or a refresher, refer to [this document](#).

Problem 1: Simple regression properties

Consider the following simple regression model:

$$Y = \beta_0 + \beta_1 x + \epsilon$$

with $\epsilon \sim \mathcal{N}(0, \sigma^2)$. Suppose we are given n observation pairs $\{(x_1, Y_1), \dots, (x_n, Y_n)\}$. Denote the fitted regression line as $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 x$. Set:

$$\begin{aligned} \text{SST} &= \sum_{i=1}^n |Y_i - \bar{Y}|^2, \quad \text{SSR} = \sum_{i=1}^n |\hat{Y}_i - \bar{Y}|^2, \quad \text{SSE} = \sum_{i=1}^n |Y_i - \hat{Y}_i|^2, \\ S_{xy} &= \sum_{i=1}^n (x_i - \bar{x})(Y_i - \bar{Y}), \quad S_{xx} = \sum_{i=1}^n |x_i - \bar{x}|^2 \end{aligned}$$

Show the following,

- (a) $\text{SST} = \text{SSR} + \text{SSE}$,
- (b) $\text{SSE} = \text{SST} - \hat{\beta}_1 S_{xy}$, and $\mathbb{E}(\text{SSE}) = (n - 2)\sigma^2$,
- (c) $\text{Cov}(\hat{\beta}_0, \hat{\beta}_1) = \frac{-\sigma^2 \sum_{i=1}^n x_i}{n \sum_{i=1}^n x_i^2 - (\sum_{i=1}^n x_i)^2}$,
- (d) $\text{Cov}(Y_i, \hat{Y}_i) = \sigma^2 \left[\frac{1}{n} + \frac{(x_i - \bar{x})^2}{S_{xx}} \right]$,
- (e) $\text{Cov}(\bar{Y}, \hat{\beta}_1) = 0$.

Problem 2: Reduced simple regression model

- (a) Consider the following simple regression model without the intercept coefficient β_0 :

$$Y = \beta_1 x + \epsilon, \epsilon \sim \mathcal{N}(0, \sigma^2)$$

Use the least squares method, show that the estimate $\hat{\beta}_1$ for β_1 is given by

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n x_i Y_i}{\sum_{i=1}^n x_i^2}$$

- (b) Consider the following simple regression model without the slope coefficient β_1 :

$$Y = \beta_0 + \epsilon, \epsilon \sim \mathcal{N}(0, \sigma^2)$$

Use the least squares method, show that the estimate $\hat{\beta}_0$ for β_0 is given by

$$\hat{\beta}_0 = \bar{Y}$$

Problem 3: Simple regression with iris dataset

The `iris` dataset contains information about the iris flowers divided into 3 sub-species: `setosa`, `virginica`, and `versicolor`. Use the command `data(iris)` to load dataset into the `R` environment.

- (a) Using the `subset` function, extract all rows from the dataset that contains information about the `versicolor` species. Store this dataframe in a variable named `iris_versicolor`.
- (b) Create a scatter plot between `Sepal.Length` and `Petal.Length` of the `iris_versicolor` dataset. Interpret the plot.
- (c) Consider the following linear model on the `iris_versicolor` dataset:

$$\text{Petal.Length} = \beta_0 + \beta_1 \cdot \text{Sepal.Length} + \epsilon, \epsilon \sim \mathcal{N}(0, \sigma^2) \quad (\text{MODEL_01})$$

Construct a linear model in `R`, and then use `summary` to view the estimated values. What can you say about β_0 ? Is it fair to consider $\beta_0 = 0$? If so, construct a second model of the form:

$$\text{Petal.Length} = \beta_1 \cdot \text{Sepal.Length} + \epsilon, \epsilon \sim \mathcal{N}(0, \sigma^2) \quad (\text{MODEL_02})$$

in `R`.

- (d) Which model is nested in another model? Using the function `anova` to test the following hypothesis:

$$\begin{cases} H_0 : \text{MODEL_02 is appropriate} \\ H_1 : \text{MODEL_01 is appropriate} \end{cases}$$

What is the p -value of this test? What is your conclusion?

- (e) Under `MODEL_02`, plot the residuals of the model against indices. What is your conclusion?
- (f) Another way to check normality is to use a so-called `qqplot` for the residuals against the normal distribution. Use the functions `qqnorm` and `qqline` in `R` to the residuals of the `MODEL_02`. What is your conclusion?
- (g) Create a scatter plot of the residuals against the fitted values. This plot is called "Residuals vs. Fitted" plot. Answer the following questions:
 - (i) Do the residuals bounce randomly around 0? If so, we can say linearity assumption is justified.
 - (ii) Do the residuals form a band? If so, the variance of the errors can assume to be equal.
 - (iii) Are there any outliers among the residuals?
- (h) Use the `predict.lm` function to make a 95% prediction of the following dataset:

$$\text{Sepal.Length} = (5.21, 5.92, 7.00, 5.94, 6.79)$$

Problem 4: Multiple regression with the Fish dataset

Download the `Fish.csv` dataset on the class website. The `Fish.csv` dataset contains multiple fish measurements of different fish species:

- **Species:** Species name of the fish.
- **Weight:** Weight of the fish, in grams.
- **Length1:** Vertical length of the fish, in cm.
- **Length2:** Diagonal length of the fish, in cm.
- **Length3:** Cross length of the fish, in cm.
- **Height:** Height of the fish, in cm.
- **Width:** Width of the fish, in cm.

In this problem, we wish to predict the width of a fish based on other measurements, that is, we consider the following linear model:

$$\text{Width} = \beta_0 + \beta_1 \cdot \text{Length1} + \beta_2 \cdot \text{Length2} + \beta_3 \cdot \text{Length3} + \beta_4 \cdot \text{Height} + \epsilon$$

where ϵ is assumed to be normally distributed: $\epsilon \sim \mathcal{N}(0, \sigma^2)$.

- Using the `pairs` function to make a pair plot between numerical features of the dataset. Base on this plot, which attributes have a strong linear relationship to `Width` ?
- Use the `lm` function in  to estimate. Which coefficient(s) can be ignored, i.e $\beta_i = 0$? Redo the analysis with this/these variable(s) excluded. We refer to this model as `MODEL_REDUCED`.
- Perform the normality check of `MODEL_REDUCED` by either plotting residuals vs indices, residuals vs fitted values or `qqplot` of the residuals. What is your conclusion ?
- Use the function `predict.lm` to construct 95% predictive intervals of the width of the following measurements:

Length1	Length2	Length3	Height
28.62	39.60	51.69	8.82
35.25	10.19	36.49	7.70
9.16	24.83	24.43	3.41
21.70	11.85	30.10	12.34
25.81	22.38	20.36	6.55

Problem 5: Cook's distance

Each year the Scottish Hill Runners Association publishes a list of hill races in Scotland for the year. In [this link to the dataset](#), this contains the distance of the race and the climb in elevation, also the record times of men and women.

We are interested in modeling record time of women based on the distance and the climb in

elevation of the race.

- (a) You may read the dataset directly from the link above. Notice that the dataset is in `.dat` format, you will need to use the `read.table` function.
- (b) Create a pair plot between `timeW`, `distance` and `climb` variables. What is your interpretation of this plot ?
- (c) Use the `lm` function to make the following linear model:

$$\text{timeW} = \beta_0 + \beta_1 \cdot \text{distance} + \beta_2 \cdot \text{climb} + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2)$$

Check the normality assumption just as previous exercises. Can you find an outlier in this dataset ?

- (d) Cook's distance can be used to measure the influential of an observation on a linear model, that is, how much the model changes when we remove that observation. The Cook's distance of the i -th observation is given by

$$D_i = \frac{e_i^2 h_i}{(p+1)s^2(1-h_i)^2}$$

Here, p is the number of explanatory variables, s^2 is the estimator of σ^2 , e_i is the i -th residual, h_i is called the **leverage** of the i -th observation, which is defined to be the i -th diagonal element of the matrix

$$\mathbf{H} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T$$

(Notice that $\mathbf{H}\mathbf{y} = \mathbf{X}\hat{\boldsymbol{\beta}} = \hat{\mathbf{y}}$).

In **R**, to compute leverage and Cook's distance for each observation, we use the `hatvalues` and `cooks.distance`, respectively.

Use these functions, plot the Cook's distance against the indices. Can you spot an unusual data point ?

- (e) Remove the unusual data point, and redo the regression analysis.

Problem 6: Leverage

Refer to the definition of the leverage in the previous exercise.

- (a) Show that, for the simple linear regression model:

$$Y = \beta_0 + \beta_1 x + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2)$$

Then, the leverage of the i -th observation is given by

$$h_i = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{\sum_{k=1}^n (x_k - \bar{x})^2}$$

- (b) Consider the following regression model:

$$Y = \beta_0 + \beta_1 x_1 + \dots + \beta_p x_p + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2)$$

Show that the leverage of the i -th observation $h_i \in [0, 1]$, and $\sum_{i=1}^n h_i = p + 1$.

Hint:

- Show that $H^2 = H$.
- You may need the following fact, without proof, that $\text{Trace}(AB) = \text{Trace}(BA)$ for any $n \times m$ matrix A , and matrix B of size $m \times n$.